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Source: Northwest Science, 94(3-4) : 286-301

Published By: Northwest Scientific Association

URL: <https://doi.org/10.3955/046.094.0306>

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Temporal Variability in Climate-Growth Response of Mountain Hemlock at Treeline in Washington and Oregon

Abstract

Climate-growth relationships in altitudinal treeline forests are particularly informative because they represent the upper limit of a species range where growth is often especially sensitive to climatic variation. Radial growth response to climatic variables typically ranges from energy limited to water limited. In the Pacific Northwest region of North America, treeline forests are typically energy limited. Results from a recent study (Marcinkowski et al. 2015) indicate a change in climate-growth relationships over time in treeline mountain hemlock (*Tsuga mertensiana*) in northern Washington state. Here, we investigate whether these same relationships hold across 700 km of the range of treeline mountain hemlock in Oregon and Washington. Using cores from trees > 200 years old, we examined temporal and spatial variability of climate-growth relationships since the early 20th century. Results indicate 1) a potential weakening of energy limitation in forest growth at treeline, 2) a nonstationary growth response to climate variables through time, and 3) different patterns of growth response to climate from north to south. Climate-growth relationships were generally similar between different aspects in the same geographic location, with some exceptions. Correlations between climate variables and radial growth differ over time (ranging from significantly positive to significantly negative), indicating the non-stationary nature of climate-growth relationships. These results identify where changes in growth-limiting variables may be occurring at treeline. Additional changes in the effects of climate change on growth at treeline are expected, and may be temporally complex.

Keywords: energy limited, mountain hemlock, subalpine, tree growth, treeline

Introduction

Climate-Growth Response at Treeline

The growth response of coniferous tree species in the Pacific Northwest region of North America to changing climate can be characterized as a gradient from energy limited to water limited (Albright and Peterson 2013). However, pure energy limitation and pure water limitation represent the extremes, and most forests fall between the two (Ettinger et al. 2011). Growth in energy-limited forests is typically constrained by temperature and length of the growing season (Ettl and Peterson 1995, Gedalof and Smith 2001a), whereas growth in

water-limited forests is typically constrained by climatic factors influencing soil water availability, such as solar radiation and evapotranspiration (Case and Peterson 2005, Littell et al. 2008).

Energy limitation is frequently associated with biogeographic and growth studies of subalpine forests, especially near treeline (Malanson et al. 2007). The altitudinal treeline represents an ecotone in which trees gradually become less dominant with increasingly unfavorable environmental conditions. A balance between abiotic climatic factors and biotic tolerance determines the survival of individual trees and the upper limit of tree growth (Arris and Eagleson 1989, Körner and Paulsen 2004). Extended duration of below-freezing temperatures and snowpack results in a truncated growing season compared to lower-elevation forests (Malanson et al. 2007), and although it may be possible for trees to be sensitive to soil moisture in some locations, high

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snowpack and a short growing season are usually more limiting. Exposure increases at treeline because of higher wind velocity, increased solar radiation, and decreased forest cover (Wieser and Tausz 2007). Individual trees that occupy this threshold between physiologically habitable and inhabitable environments are especially sensitive to growth-limiting factors (Brubaker 1986, Kipfmüller and Salzer 2010), and even minor changes in climate (especially extremes) can influence annual growth by altering the rate of physiological processes (Fritts 1976, Wieser 2007, Bunn et al. 2011, Salzer et al. 2014).

Spatial and temporal variability at relatively small scales is especially influential on plant growth in the subalpine environment, because microsite extremes (e.g., temperature) are intensified in the absence of a closed forest canopy (Wieser 2007). Differences in radiation inputs between north- and south-facing slopes result in differences in soil and air temperatures, snow cover duration, soil moisture, and the resulting distribution of vegetation (Malanson 2001). For example, in the Pacific Northwest, the elevation of treeline is typically 150 m higher on southerly aspects than nearby northerly aspects (Franklin and Dyrness 1973). In the subalpine environment, even slight differences in topographic position may result in differences in growth responses to climate (Villalba et al. 1994, Bunn et al. 2011, Tran et al. 2017), including a shift from energy limitation to water limitation from northerly to southerly sites (respectively) within the same approximate elevation band (Leonelli et al. 2009).

Annual average temperature in the Pacific Northwest has increased 0.8 °C during the 20th century, and is expected to continue to increase an average of 0.3 °C per decade during the 21st century (Abatzoglou et al. 2014). Increased temperatures have resulted in reduced precipitation in the Cascade Range (Luce et al. 2013), reduced snowpack (Mote et al. 2018), a longer freeze-free season, and an increase in climatic water deficits (Abatzoglou et al. 2014). Seasonal projections include higher temperatures and less precipitation in the summer, and higher temperatures and slightly more precipitation, mostly falling as rain

(especially below 2,000 m elevation), in the winter (Mote and Salathé 2010). These projected changes in climate have the potential to alter environmental conditions that limit tree growth near treeline.

Mountain Hemlock Climate-Growth Response Studies

Mountain hemlock (*Tsuga mertensiana* (Bong.) Carrière) is a dominant subalpine tree species throughout the west side of the Cascade Range and Olympic Mountains and has been the subject of previous climate-growth investigations and climatic reconstructions (Graumlich and Brubaker 1986; Taylor 1995; Smith and Laroque 1998; Gedalof and Smith 2001a, b; Peterson and Peterson 2001; Laroque and Smith 2003, 2005; Marcinkowski et al. 2015). Mountain hemlock growth exemplifies the gradient from energy limited to water limited (Peterson and Peterson 2001).

At high-elevation sites, growth is typically correlated with variables that influence growing season length, indicating that energy limits growth. Growth is negatively correlated with winter precipitation and snowpack and previous-year summer temperatures (Smith and Laroque 1998, Gedalof and Smith 2001a, Peterson and Peterson 2001). Growth is positively correlated with current-year summer temperatures (Smith and Laroque 1998, Gedalof and Smith 2001a, Peterson and Peterson 2001) and with winter Pacific Decadal Oscillation (PDO) index (where a positive PDO index is generally associated with climate that is warmer and drier than average in the eastern Pacific region) (Gedalof and Smith 2001b, Peterson and Peterson 2001, Marcinkowski et al. 2015). Growth is also positively correlated with spring El Niño Southern Oscillation (ENSO) index (where a positive ENSO index is generally associated with climate that is warmer and drier than average in the eastern Pacific region) (Marcinkowski et al. 2015).

At low-elevation sites, mountain hemlock growth is typically water limited. Growth is negatively correlated with previous-year summer temperature and with winter PDO index (Peterson and Peterson 2001). Growth is positively correlated with previous-year summer precipitation (Smith

and Laroque 1998, Gedalof and Smith 2001a, Peterson and Peterson 2001, Marcinkowski et al. 2015).

Peterson and Peterson (2001) laid a foundation for investigating changes in growth response to climate throughout the range of mountain hemlock populations in the Pacific Northwest. Mountain hemlock in Washington and northern Oregon displayed a climate-growth relationship typical of an energy-limited system at high elevations and a water-limited system at low elevations. At lower-latitude sites in southern Oregon, climate-growth correlations suggest that a different set of climate variables controls growth. Growth was negatively correlated with spring snowpack depth, suggesting energy limitation, but negatively correlated with previous-year summer temperature and previous-year August precipitation, suggesting water limitation. In summary, these results demonstrated that mountain hemlock growth was influenced by climatic factors on an annual to decadal scale, exhibiting energy-limited to water-limited growth responses along elevation and latitudinal gradients.

Effects of Increased Temperature on Growth

In energy-limited subalpine forests, if climate change is manifest as warmer temperatures, decreased snowpack, and increased precipitation falling as rain, a longer growing season and increased productivity would be expected (Latta et al. 2010). Since the early 20th century, several coniferous species showed an increase in growth with increasing temperatures in the Northern Hemisphere:

- Washington: western hemlock (*Tsuga heterophylla* (Raf.) Sarg.), Pacific silver fir (*Abies amabilis* Douglas ex J. Forbes), and mountain hemlock (Graumlich et al. 1989);
- California and Nevada: bristlecone pine (*Pinus longaeva* D. K. Bailey) (Salzer et al. 2014) and limber pine (*Pinus flexilis* James) (La Marche et al. 1984);
- California: lodgepole pine (*Pinus contorta* Douglas ex Loudon var. *murrayana* (Balf.) Engelm.), whitebark pine (*Pinus albicaulis*

Engelm.) (Peterson et al. 1990), and limber pine (Millar et al. 2015);

- California: whitebark pine and western white pine (*Pinus monticola* Douglas ex D. Don) (Millar et al. 2004);
- Western United States and Canada: chronologies composed mainly of ponderosa pine (*Pinus ponderosa* Lawson & C. Lawson), Engelmann spruce (*Picea engelmannii* Parry ex Engelm.), Douglas-fir (*Pseudotsuga menziesii* (Mirb.) Franco) (McKenzie et al. 2001), and white spruce (*Picea glauca* (Moench) Voss) (Porter and Pisaric 2011);
- Europe: Scots pine (*Pinus sylvestris* L.), mugo pine (*Pinus mugo* Turra), Bosnian pine (*Pinus leucodermis* Antoine), Austrian pine (*Pinus nigra* Arnold), Balkan pine (*Pinus peuce* Griseb.), silver fir (*Abies alba* Mill.), and Norway spruce (*Picea abies* (L.) Karst) (Briffa 1992).

In addition, studies across the Northern Hemisphere, sampling different tree species, have documented a decreased sensitivity to temperature:

- Review of many studies of various species at high latitude and high elevation (D'Arrigo et al. 2008);
- Alaska and northwestern Canada: white spruce at latitudinal and altitudinal treeline (Jacoby and D'Arrigo 1995, Porter and Pisaric 2011);
- Italian Alps: European larch (*Larix decidua* Mill.) at altitudinal treeline (Coppola et al. 2012);
- Siberia: Dahurian larch (*Larix gmelinii* (Rupr.) Rupr.) at latitudinal treeline (Jacoby et al. 2000);
- Kyrgyzstan: Juniper species (*Juniperus* spp. L.) at altitudinal treeline (Esper et al. 2003);
- California: bristlecone pine near altitudinal treeline (Salzer et al. 2014).

A weakening in sensitivity to temperature was detected in a large-scale study of boreal forest growth chronologies since the early 1960s (Briffa et al. 1998). Tree growth increased with the high temperatures recorded for the 1930s and 1940s, decreased during the cooler 1950s, and did not track the continuous increase in temperatures that began in the 1970s. Decreased sensitivity to increased temperature was attributed to summer

drought, increased competition with other plants, and perhaps other factors.

Peterson and Peterson (2001) found that growth at high-elevation sites in the Cascade Range of Washington and northern Oregon was positively correlated with summer temperature and negatively correlated with winter precipitation and 01 April snow-water equivalent (SWE). However, D'Arrigo et al. (2004) found a weakening in the energy-limited signal at high-latitude treeline sites. In the Cascade Range of Washington, Marcinkowski et al. (2015) found climate-growth results similar to those of D'Arrigo et al. (2004). At sites sampled by Marcinkowski et al. (2015), growth was not correlated with winter precipitation or April 01 SWE, and was correlated with spring temperature, and climate-growth relationships were non-stationary over time. The difference among the results of these three similar studies suggests that a change may have occurred in limiting factors and climate-growth relationships over time.

Potential for New Insights

Research on mountain hemlock growth response to climatic factors in the Pacific Northwest (Peterson and Peterson 2001), and recent insight into potential changes in limiting factors (Marcinkowski et al. 2015), need to be reconciled with a broader contemporary sample. Treeline forests are ideal for detecting climatic effects because of 1) climatic sensitivities of trees living at the farthest extent of their distribution (Malanson et al. 2007), 2) accelerated and amplified changes in temperature and snowpack documented at high-elevation sites (Mote et al. 2018), 3) their ecological importance as ecotones (EDW 2015), and 4) established climate-growth relationships that serve as a proxy for reconstructing past climates and modeling future climates (Jacoby and D'Arrigo et al. 1995). Quantifying changes in these relationships will improve the accuracy of modeling the effects of climatic variability and change on future forest growth.

In this study, we investigated climate-growth correlations for subalpine mountain hemlock on north- and south-facing aspects on the west side of the Cascade Range and Olympic Mountains in

Washington and Oregon. We used a multidecadal analysis period to explore 1) variability throughout time in climate-growth correlations, 2) variability between aspects for climate-growth relationships, and 3) variability among populations from northern Washington to southern Oregon.

Methods

Study Area

Sampling was conducted in subalpine forests on the west side of the Cascade Range in Washington and Oregon, and in the Olympic Mountains of Washington (Figure 1). Subalpine forests are found at the highest elevations of these mountains where temperature and snowpack are at the limit of what is needed to support the physiological function of trees (photosynthesis, respiration, etc.). This results in a gradual decrease of continuous forest into discontinuous forest (e.g., tree islands), as conditions become less favorable with increasing elevation (Malanson et al. 2007). The Olympic Mountains and the west side of the Cascade Range in Washington and northern Oregon have a maritime climate that is strongly influenced by the Pacific Ocean. Winters are wet and cold, and summers are warm and dry. The subalpine area is an extreme extension of the maritime climate, where high amounts of precipitation fall as snow early in autumn, and low temperatures maintain a deep snowpack that can persist into the summer months.

Nine high-elevation locations were sampled in the summer of 2016. At each location, a dominant north and south aspect was sampled, for a total of 18 sites. Sites were chosen along a latitudinal transect from northern Washington to southern Oregon, spanning the range of mountain hemlock populations in the Cascade Range and Olympic Mountains. (Table 1). All sites were above 1,300 m in elevation. It should be noted that the Baker site is approximately 11 km to the northeast of Mt. Baker.

At each site, sampling was conducted by hiking above the treeline, and then slowly dropping elevation until 20 suitable trees were located on the respective aspect, and cores were collected



Figure 1. Locations where mountain hemlock was sampled. Sites are abbreviated throughout manuscript to Baker, Olympic, Minotaur, Rainier, Adams, Hood, Jefferson, Bachelor, and Crater Lake. Map created by Robert Norheim.

(one core per tree). This method ensured that the highest-elevation trees were sampled at each site. Suitable trees were defined as > 200 years old (to eliminate juvenile growth trends), dominant or codominant (to eliminate strong signals from competition), and with no obvious structural flaws (e.g., broken tops, insect damage, or fire damage). The location of all sampled trees was recorded using a hand-held GPS, and the height and diameter at breast height (dbh) were measured.

Chronology Construction

Trees were cored at breast height (1.37 m), using an increment borer on the cross-slope position whenever possible to avoid compression wood (Fritts 1976). Tree cores were stored in paper

straws for transport prior to being mounted in grooved boards and sanded with progressively finer grades of sand paper. Tree rings were measured to the nearest 0.001 mm with a measuring stage (Velmex Inc., Bloomfield, New York), Measure J2x software (version V4.2 VoorTech Consulting, Holderness, New Hampshire), and an Olympus DF PLAN 1.5x measuring scope. Tree rings were crossdated using standard dendrochronological techniques, and crossdating was applied to each series. (Stokes and Smiley 1968). Crossdating was substantiated by creating mean chronologies for each site and calculating time-shifted correlation coefficients for each core with the program COFECHA (Holmes 1983).

Each series was detrended with the R (R Core Team, 2016) package dplR (Bunn et al. 2017) to remove trends in growth from age-related

biological processes. A cubic spline with a 50% cutoff of 150 years was used for detrending in order to maximize annual and decadal variability without washing out trends occurring over longer time spans. Other cutoffs were investigated, but the results of the detrending were similar for all of them, so a cutoff of 150 years was chosen for comparability to similar studies in the region (Marcinkowski et al. 2015, Gedalof and Smith 2001a, Peterson and Peterson 2001).

The detrended series were standardized into a unitless index by dividing the ring-width measurement by the value of the fitted curve. In order to maximize the climate-growth signal, the individual series were prewhitened with an autoregressive model to remove autocorrelation. Chronologies

TABLE 1. Summary of site and core characteristics.

Site	Mean elevation (m) (SD)	Longitude (dec. deg.)	Latitude (dec. deg.)	Mean dbh (cm) (SD)	Date range of cores	Median age (yr)	Mean age (yr)
Baker N	1,256 (7.3)	-121.67	48.86	86.1 (15.5)	1473–2015	370	365
Baker S	1,507 (59.3)	-121.82	48.85	73.3 (19.0)	1542–2015	317	329
Olympic N	1,348 (49.3)	-123.78	47.92	61.8 (18.4)	1445–2015	284	309
Olympic S	1,580 (28.6)	-123.77	47.90	49.7 (9.7)	1602–2015	296	312
Minotaur N	1,663 (159.9)	-121.03	47.85	59.3 (13.5)	1547–2015	267	293
Minotaur S	1,719 (66.5)	-121.03	47.83	64.1 (9.3)	1698–2015	223	236
Rainier N	1,746 (65.4)	-121.58	46.92	77.0 (12.1)	1639–2015	252	260
Rainier S	1,539 (24.7)	-121.72	46.77	75.0 (12.4)	1609–2015	243	258
Adams N	1,852 (17.6)	-121.54	46.23	75.0 (12.7)	1631–2015	254	263
Adams S	1,904 (22.3)	-121.49	46.15	70.6 (10.9)	1607–2015	253	265
Hood N	1,843 (10.1)	-121.70	45.41	61.2 (14.5)	1595–2015	293	300
Hood S	1,841 (21.1)	-121.72	45.33	66.0 (11.0)	1609–2015	317	313
Jefferson N	1,842 (41.1)	-121.80	44.70	79.5 (18.2)	1593–2015	273	295
Jefferson S	1,823 (18.4)	-121.82	44.61	84.2 (13.2)	1563–2015	285	299
Bachelor N	2,068 (15.6)	-121.68	44.00	85.4 (12.1)	1636–2015	300	304
Bachelor S	2,226 (11.2)	-121.96	43.96	80.1 (13.9)	1633–2015	311	309
Crater Lake N	2,121 (25.1)	-122.11	42.98	105.2 (16.8)	1554–2015	308	326
Crater Lake S	2,049 (10.6)	-122.11	42.84	95.3 (19.5)	1601–2015	280	290

were generated for each site by averaging all of the residual ring-width series for each site using a Tukey’s biweight robust mean. A correlation matrix was created from the residual site chronologies using Pearson’s product-moment correlation to assess the similarities in growth between sites.

Climate Data

We determined which seasonal climate variables would be most appropriate for analysis based on similar previous studies and assessment of monthly correlations (Peterson and Peterson 2001, Marcinkowski et al. 2015). The following climatic variables are included in this discussion: spring (April to May) temperature, summer (June to August) temperature, winter (November to March) precipitation, and summer (June to August) precipitation. The following climate variables were also analyzed but were not found to be consistently correlated with tree growth: summer Palmer Drought Severity Index (PDSI), winter ENSO index, winter PDO index, previous growth-year winter precipitation, previous growth-year summer temperature, and previous growth-year spring temperature.

Monthly climate data for temperature and precipitation, spanning 1896 to 2015, were obtained from the Climate Divisional Dataset (Vose et al. 2014). Divisional data are aggregations of weather station data considered to be coherent within a particular geographic area. There are 10 climate divisions in the state of Washington (344 climate divisions in the continental United States). For each climate division, monthly station temperature and precipitation values are computed from daily observations. Divisional values are weighted by area to compute statewide values and are weighted by area to compute regional values (NOAA 2017). Divisional data were used because the compilation of individual station data within a climatic region better represents the overall regional climate, and has a longer record than a single station. The 18 sites fell within the boundaries of 8 different climate divisions. Trees sampled at the north aspect site of Minotaur Lake fell between two divisions, so the division containing more samples was used. PDO index data for 1901 to 2015 were obtained from the University of Washington Joint Institute for the Study of the Atmosphere and Ocean climate data archives (JISAO 2017). The Niño region 3.4 Index (ENSO)

TABLE 2. Cross correlations are shown for the climate variables that had the highest significance with growth across all sites.
^aIndicates significant correlation ($P < 0.05$); PY, previous year; temp, temperature; precip, precipitation.

	Spring temp	PY spring temp	Summer temp	PY summer temp	Winter precip	PY winter precip	Summer precip
Spring temp							
PY spring temp	0.02						
Summer temp	0.27 ^a	0.14					
PY summer temp	0.17	0.26 ^a	0.26				
Winter precip	-0.18 ^a	-0.26 ^a	-0.11 ^a	0.02			
PY winter precip	-0.05	-0.18	0	-0.11	-0.06		
Summer precip	-0.09	0.04	-0.36 ^a	0.05	0	0	

for 1870 to 2015 was obtained from the National Oceanic and Atmospheric Administration (NOAA 2018). Respective monthly climate variables were averaged to create seasonal climate variables for spring, summer, and winter.

Climate-Growth Correlations

To determine the significance of climate variables for mountain hemlock growth, monthly Pearson product-moment correlations ($\alpha = 0.05$) were calculated between site chronologies and monthly climate variables. The chronologies were shortened to the 120-year length of the climate records, spanning 1896 to 2015, and then divided into 30-year climate normal periods. We designated the climate normal periods 1986 to 2015, 1956 to 1985, 1926 to 1955, and 1896 to 1925 to match the duration of the available climate data and the end date of the tree-growth data. These are slightly different than what are typically called US Climate Normals, but climate normal periods are used flexibly in scientific analyses. Pearson product-moment correlations ($\alpha = 0.05$) and confidence intervals were calculated between chronologies and climate records for each climate normal. Climate variables are often associated with one another, so correlations between climate variables were calculated to ensure that we understood any potential connections (Table 2).

Results and Discussion

Chronology Descriptions

Chronology statistics were calculated to infer appropriateness for use in climate-growth analysis

based on widely accepted ranges (Table 3). Expressed population signal (EPS) measures how similar a chronology is to an ideal hypothetical chronology. A general guide for acceptable EPS values is ≥ 0.85 (Wigley et al. 1984); values in this study ranged from 0.864 to 0.929 (Buras [2017] claims this threshold is arbitrary). Sample size was 20 for the period of the chronologies used for climate analysis (1896 to 2015). Mean interseries correlation measures the similarity of growth response between trees at a site. Values ranged from 0.497 to 0.684, which suggests a similar response to climatic variability between the chronologies. Subsequent discussion of the results is based primarily on the four climate variables (winter precipitation, spring temperature, summer temperature, and summer precipitation) that had the highest correlations with growth chronologies.

Correlation Analysis

Correlations between chronologies at north vs. south aspects of the same site are high, with correlations at southern latitude higher than at northern latitude (Table 4). Correlations between Oregon sites are generally higher (highest correlations with other Oregon sites range from 0.78 to 0.86) than correlations between Washington sites (highest correlations with other Washington sites range from 0.64 to 0.79). All sites except three (Minotaur N, Adams N, and Adams S) have the highest correlations with either the opposite aspect at the same site (Baker N, Minotaur S, Rainier N/S, Jefferson N/S, Bachelor N/S, and Crater Lake N/S), or with the same aspect at an adjacent site (Baker S, Olympic N/S, Hood N/S).

TABLE 3. Basic dendrochronological statistics were calculated for the 18 chronologies compiled in the study. See text for detail on specific statistics. An N after the site name indicates north aspect; an S indicates south aspect. EPS values are for 120 years of the series used in analysis.

Site	<i>n</i>	Series mean first-order autocorrelation	Mean interseries correlation	Autoregressive order	Expressed population signal (EPS)	Mean sensitivity
Baker N	20	0.719	0.530	4	0.892	0.205
Baker S	20	0.665	0.584	4	0.884	0.256
Olympic N	20	0.708	0.502	15	0.864	0.241
Olympic S	20	0.664	0.573	5	0.911	0.259
Minotaur N	20	0.682	0.497	2	0.879	0.230
Minotaur S	20	0.686	0.546	3	0.865	0.249
Rainier N	20	0.696	0.551	11	0.899	0.207
Rainier S	20	0.753	0.591	5	0.916	0.205
Adams N	20	0.631	0.662	2	0.924	0.247
Adams S	20	0.691	0.606	5	0.915	0.243
Hood N	20	0.573	0.589	2	0.893	0.267
Hood S	20	0.629	0.641	6	0.932	0.260
Jefferson N	20	0.624	0.602	16	0.903	0.252
Jefferson S	20	0.612	0.627	4	0.920	0.267
Bachelor N	20	0.709	0.624	4	0.897	0.244
Bachelor S	20	0.710	0.605	2	0.892	0.239
Crater Lake N	20	0.621	0.684	4	0.929	0.262
Crater Lake S	20	0.651	0.676	4	0.897	0.247

There is considerable variability in mountain hemlock growth over time: site correlations are non-stationary over the analysis time period, with correlation values varying and moving in and out of significance between climate normals (Table 5). For example, Jefferson S is barely negatively correlated with winter precipitation for the climate normal period 1896 to 1925, becomes increasingly negatively correlated for 1926 to 1955, followed by significantly negatively correlated for 1956 to 1985, and less negatively correlated for 1986 to 2015.

Growth-Limiting Factors

Previous studies have demonstrated that high-elevation mountain hemlock growth is controlled by climatic variables that influence length of the growing season (and therefore energy limitation) (Peterson and Peterson 2001, Albright and Peterson 2013), and these relationships are non-stationary through time (Driscoll et al. 2005, Lloyd and Fastie 2002, Marcinkowski et al. 2015, Sherriff et al. 2017). Climate-growth relationships for mountain hemlock in our study 1) are non-stationary

over time, thus concurring with previous studies, 2) differ by location, with some specific geographic patterns, and 3) mostly indicate that growth at treeline is energy limited but that this relationship may have weakened in some locations in Washington and Oregon (but does not indicate that growth is water limited).

Growth is at least partially energy limited at these sites. Winter precipitation is overall negatively correlated with growth (Figure 2), and has similar correlations for the northernmost sites; Baker N and Olympic N are similarly correlated throughout climate normal periods (Baker S and Olympic S are similarly correlated). There are more negative correlations with winter precipitation in the two later climate normal periods than the two earlier. However, the strength of the correlation between winter precipitation and growth differs among climate normal periods (e.g., growth is significantly correlated with winter precipitation for 1956 to 1985 at more sites than for 1986 to 2015).

During these same climate normal periods, more sites became significantly positively correlated with spring temperature (Figure 3). This recent

[illegible]

TABLE 5. Major inferences from results by climate variable.

Spring Temperature
• Overall, positively correlated.
• Overall, no discernible difference between aspect and latitude.
• Correlations are variable by site, latitude, and aspect.
• Fewer sites are significantly correlated with spring temperature for 1926–1955 and 1956–1985.
• Southern latitude sites are more consistently significantly correlated (although no southern latitude sites are significantly correlated during 1956–1985).
Summer Precipitation
• Few correlations are significant.
• Correlations are predominantly negative or static.
Winter Precipitation
• Overall, correlations between growth and winter precipitation are negative.
• No significant correlations for 1896–1925.
• All sites except Bachelor are significantly negatively correlated with winter. precipitation for 1956–1985.
• Correlations are variable through time, by latitude, and by aspect.
Summer Temperature
• Overall, correlations between growth and summer temperature are positive.
• 1986–2015 has far more significant correlations than other climate normal periods.
• 1896–1925 has no significant correlations.
Overall
• Correlations are variable by climate normal, site and aspect.
• Spring temperature, winter precipitation, and summer temperature have the most significant correlations with growth. Summer precipitation has fewer significant correlations.
• Correlations are variable throughout time periods, by aspect, and latitude.
• The Olympic sites have the fewest significant correlations.
• Climate variable with the most significant correlations can change by climate normal.
• Some sites have a greater number of significant correlations overall.

change is similar to results from Marcinkowski et al. (2015) who posited that a decrease in sensitivity to the length of the growing season, and an increase in sensitivity to spring temperature, may suggest a shift from energy limitation to an intermediate response between energy and water limitation. However, similar fluctuations in and

out of significance are observable for 1896 to 1925 and 1926 to 1955. In addition, spring temperature cannot be considered as the only variable affecting energy and an “early spring,” because the growing season for mountain hemlock is also affected by depth of spring snowpack (Peterson and Peterson 2001). Summer temperature is positively correlated with growth (Figure 4), and summer precipitation is overall negatively correlated (Figure 5), which implies that mountain hemlock at these sites are not water limited. Growth responses along a gradient from energy to water limitation may differ as a function of climatic variability over time.

Effects of Aspect on Growth

Aspect did not greatly affect climate-growth relationships of mountain hemlock. Aspect is rarely considered as a mechanism controlling growth in the subalpine ecosystem, because the dominance of snow as a limiting factor is thought to diminish the importance of topographic position of trees (Peterson 1998), although some studies have shown that aspect modifies local conditions enough to affect growth at treeline (Villalba et al. 1994, Bunn et al. 2011). We found that there are generally similar climate-growth relationships between the two aspects, however the Minotaur Lake and Mt. Rainier sites, which receive similar amounts of winter precipitation, have a distinctly different pattern by aspect.

Overall, different aspect sites at the same location follow similar patterns throughout time. When aspects have a different response to climate, they have periods of convergence; for example, when winter precipitation and summer temperature for the north and south aspect sites at Mt. Baker have distinctly different correlations throughout most of the analysis, but aspects for both variables converge to nearly identical correlations at various points. The lack of a clear trend in climate-growth response by aspect indicates that other factors (especially snow) overwhelmingly influence growth in the subalpine ecosystem, and topographic aspect has less influence on growth than other factors.

Variability in climate-growth response among time periods implies that limiting factors for high-

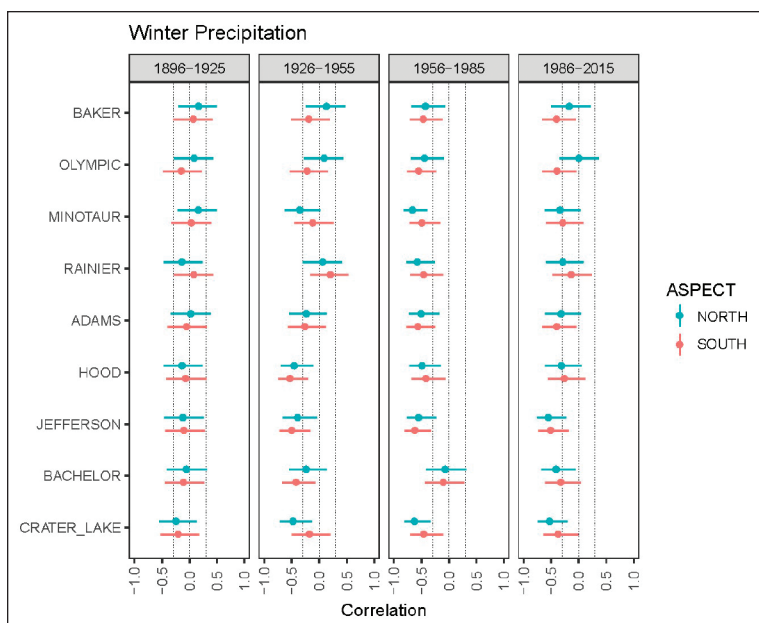


Figure 2. Pearson product-moment correlations by climate normal time period. Bars represent 95% confidence interval, outer dashed lines represent significance.

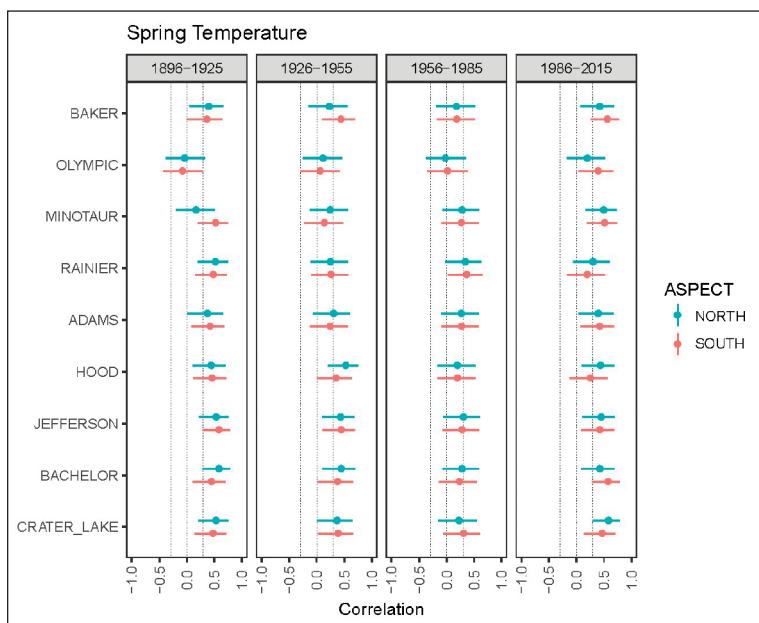


Figure 3. Pearson product-moment correlations by climate normal time period. Bars represent 95% confidence interval, outer dashed lines represent significance.

elevation mountain hemlock growth have shifted over time (Fritts 1976, Trindade et al. 2011). For example, for 1896 to 1925, spring temperature is significantly correlated with the most sites, which suggests that growth is energy limited. The climate normal 1926 to 1955 is inconsistent because all four variables are significant for different sites. During the 1956 to 1985 climate normal, winter precipitation is significantly negatively correlated with all sites except Bachelor, again suggesting that growth is energy limited. The climate normal 1986 to 2015 is a combination of significant correlations between growth and spring temperature, summer temperature, and winter precipitation, depending on the site. This change from earlier in the time series suggests that the effects of climate on growth may have changed.

It also appears that growth at some sites is more sensitive in general, and some sites may be more sensitive to specific variables during certain climate normals. For example, Olympic N and S, and Minotaur N are not significantly correlated with any of the variables for 1896 to 1925, and Jefferson S is significantly correlated with all four climate variables for 1986 to 2015. In addition, the Olympic sites have fewer significant correlations than any of the other sites. It is difficult to infer causation

for differences among sites, although local factors not captured by climate data, soil characteristics, and genetic characteristics of trees might account for some of the differences.

Conclusions

This study used a spatially extensive, contemporary sample to investigate a potential change in growth-limiting climate factors for a large portion of the range of high-elevation mountain hemlock. Mountain hemlock climate-growth relationships at treeline differed over time and to some extent over space. Recent changes in the effects of winter precipitation and spring temperature on growth suggest that the strength of energy limitation on growth may have weakened slightly. Climate-growth relationships are typically similar across latitude, where the highest latitude sites share a similar signal over time, the lowest latitude sites share a similar signal over time, and the responses of the mid-latitude sites are less consistent.

Mountain hemlock is a widespread and ecologically important species across montane regions of the Pacific Northwest, and accurate quantification of its growth and productivity are important for understanding ecosystem function (Blanco et al. 2017). Empirical results from this study can be used to project productivity and

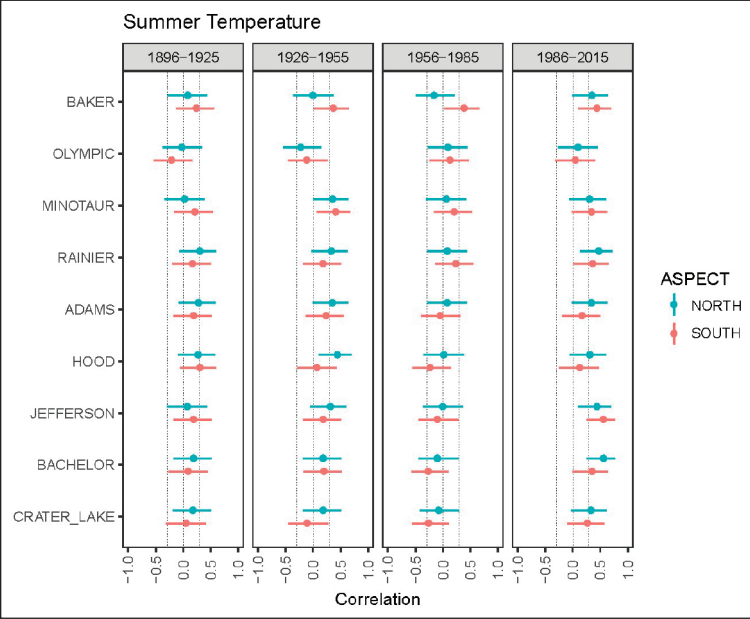


Figure 4. Pearson product-moment correlations by climate normal time period. Bars represent 95% confidence interval, outer dashed lines represent significance.

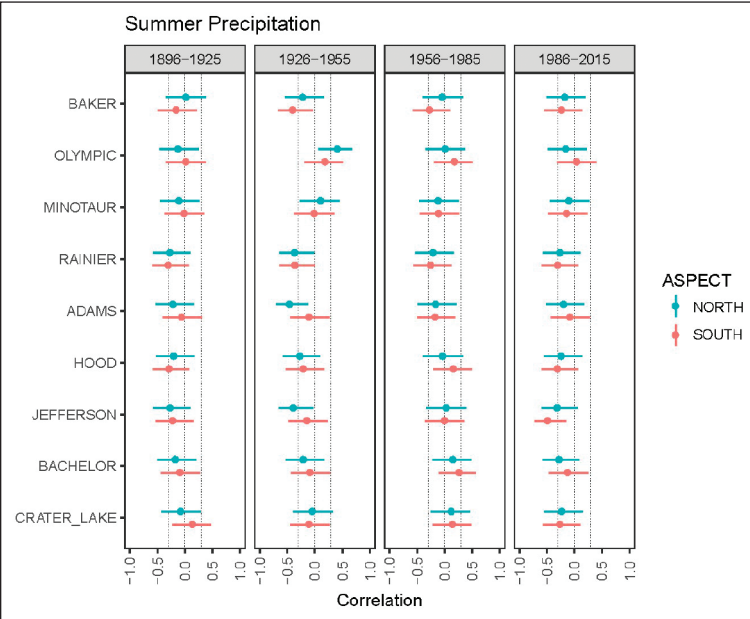


Figure 5. Pearson product-moment correlations by climate normal time period. Bars represent 95% confidence interval, outer dashed lines represent significance.

carbon storage for mountain hemlock in response to a warmer climate (Laroque and Smith 2003, 2005). The results can also inform resource managers about current conditions in subalpine forests and how they might change in the future, which is relevant for local resource planning and project management.

Variability of mountain hemlock populations in radial growth response to climate can provide input for risk management and prioritization of monitoring activities. In the future, a subset of sites could be sampled to quantify climate-growth relationships as a component of a larger monitoring program. In addition, the portion of mountain hemlock geographic distribution not included in this study could be sampled to determine if results from this study are consistent across the entire species range. The data collected in this study could also be used in an expanded analysis to further investigate if mountain hemlock on different slope aspects have significantly different climate-growth relationships, if those relationships vary through time, and if there is a difference in sensitivity to climate variables through time. Finally, this study could be conducted again in 20–30 years to determine how growth has continued to respond to climatic variability and change.

It is tempting to make generalizations from limited data on climate-growth relationships about

how a species will respond to future changes in climate. Indeed, vegetation models often simulate responses for individual species across large land areas without regard to significant sources of variability (Gustafson et al. 2016). The variable temporal response observed across the geographic domain of this analysis indicates that simple generalizations may be inaccurate. Assuming that growth is a component of fitness (Clark et al. 2011), flexible climate-growth relationships in mountain hemlock will probably ensure the persistence of this species in subalpine ecosystems in a warmer climate, at least in the absence of severe disturbances (e.g., crown fire; Dickman and Cook 1989) and large-scale regeneration failure.

Acknowledgments

We appreciate discussions on research design and analysis with Don McKenzie, Maureen Kennedy, and Kailey Marcinkowski. Comments from two anonymous reviewers greatly improved the manuscript. With thank Mike Tjoelker for assistance with fieldwork, and Rosemary Pazdral for assistance with R programming. This work was supported by funding from the US Forest Service Pacific Northwest Research Station, US Forest Service Office of Sustainability and Climate, and a fellowship from the University of Washington School of Environmental and Forest Sciences.

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Received 12 November 2019

Accepted 26 August 2020